HOMEWORK 5
NUMERICAL METHODS
FALL 2025

DUE FRIDAY, OCT 31

Instructions: Complete the following problems.

Prob 1. Provide a proof for the following statement: If points $x_0, x_1, ..., x_n$ are distinct, then for arbitrary real values $y_0, y_1, ..., y_n$, there is a unique polynomial $P_n(x)$ of degree at most n such that $P_n(x_i) = y_i$ for $0 \leq i \leq n$.
Hint You are allowed to assume existence since we have already worked with several forms of interpolating polynomials. Here we are mainly focusing in the uniqueness of the interpolating polynomial.

Prob 2. Use Lagrange interpolation process to obtain a polynomial of least degree that assumes these values

x	0	2	3	4
y	7	11	28	63

Prob 3. Continuation of Prob 2 - repeat the process as in Prob 2 but now find the Newton form of the interpolating polynomial using divided differences. Comparing the results of Prob2 and Prob3, is there any contradiction to the uniqueness theorem for polynomial interpolation? Defend your response.

Prob 4. There exists a unique polynomial $p(x)$ of degree 2 or less such that $p(0) = 0$, $p(1) = 1$, and $p'(\alpha) = 2$ for any value of α belonging to the interval $[0, 1]$, except one value of α, let us call it α_0. Determine α_0 and construct the polynomial for $\alpha \neq \alpha_0$.

Prob 5. Show that if a function g interpolates the function f at $x_0, x_1, ..., x_{n-1}$ and h interpolates f at $x_1, x_2, ..., x_n$, then

$$g(x) + \frac{x_0 - x}{x_n - x_0}[g(x) - h(x)]$$

interpolates f at $x_0, x_1, ..., x_n$.

Prob 6. Show that the trapezoid rule for numerical integration results from approximating f by a first degree-spline S and then using

$$\int_a^b f(x)dx \approx \int_a^b S(x)dx.$$

Computer Prob 1. Consider the **Runge** function given by

$$f(x) = \frac{1}{(1+x^2)}$$

perform the following tasks:

a) Construct a first degree spline using 21, 41 and 81 equally spaced points as your knots on the interval $[-5,5]$. Plot $S_{21}(x)$, $S_{41}(x)$ and $S_{81}(x)$ in a single figure. Report your findings.

b) Use 21 ChebyShev nodes and label your polynomial as $G_{20}(x)$. Using those same ChebyShev nodes are your knots, construct a 1st degree spline $\tilde{S}_{21}(x)$. Plot the newly formed 1st degree spline and $G_{20}(x)$. Report your findings.

Note1: The above tasks are to be done with the codes developed in class: *Spline1.* **Note2:** notation-wise $(n+1)$ distinct points generate an interpolating polynomial of degree n, $P_n(x)$. $(n+1)$ knots generate n sub-intervals each containing a polynomial of degree 1, and we refer to the spline as $S_{n+1}(x)$.

Important The programming should be done in terms of a Script and a function file in Matlab. Your submission should include a folder named "YourLastName" that contains 1) a .txt file with basic instructions/description of your script.m and your function.m file 2) script.m file and 3) function.m file. It is important to compress your folder before submitting it on Canvas. Try something like Winzip or Winrar. In addition to the compressed folder with all content, you also need to submit a .pdf file which contains your homework problems in project format.

*****Please follow instructions carefully - submit on time/no late acceptance *****
